waterX: Smart Plant Water Management System

Requirements and Database Design

October 15, 2026

Contents

1 Introduction **1**

2 Functional Requirements **1**

3 Non-Functional Requirements **2**

4 Database Design **3**

4.1 Overview 3

4.2 ER Diagram 3

4.3 Relationships 3

4.4 Data Dictionary 3

4.5 Indexes and Views 9

1 Introduction

waterX is a software-simulated smart irrigation system. It monitors plant conditions, decides when to water using schedules, soil moisture, tank level and safety rules, and uses machine learning to predict future soil moisture.

Research question. Can AI-assisted watering reduce water consumption compared with fixed-schedule watering without negatively affecting plant growth?

This document lists the functional and non-functional requirements and describes the SQLite database design. Priorities use the MoSCoW scale (*Must*, *Should*, *Could*).

2 Functional Requirements

ID	Requirement	Priority
FR-01	The user can add, edit and deactivate plants and assign each plant a plant profile.	Must
FR-02	The user can configure each profile's watering settings: minimum, maximum and optimal moisture, water per cycle, maximum cycles per day, maximum pump run time and an optional fixed schedule time.	Must
FR-03	The simulator generates hourly weather (temperature, humidity, sunlight, rain) and soil moisture per plant, modelling evaporation, plant uptake, watering and rain.	Must

ID	Requirement	Priority
FR-04	The simulator adds sensor noise and faults (dropouts, spikes) and flags faulty readings.	Should
FR-05	The fixed-schedule strategy waters a set amount at a set time every day (baseline for the experiment).	Must
FR-06	The rule-based strategy waters a plant when its moisture falls below the profile threshold.	Must
FR-07	Safety rules override every strategy: no watering when the tank is empty or the sensor is faulty, and limits on pump run time and waterings per day.	Must
FR-08	The AI-assisted strategy uses predicted moisture to water before moisture drops below the threshold.	Must
FR-09	The system stores every sensor reading, weather record, watering event and tank level in the database, linked to a simulation run.	Must
FR-10	The system raises alerts (low tank, too dry, too wet, sensor fault, pump limit, predicted dry) with a severity level.	Should
FR-11	The system trains and evaluates moisture-prediction models using a time-based split, reports MAE and RMSE, compares them with a naive baseline, and stores model details and predictions.	Must
FR-12	The system records plant growth (height, leaf count, biomass, stress level) and compares it with watering history.	Should
FR-13	The dashboard shows plant status, moisture charts, tank level, water consumption and alerts.	Must
FR-14	The dashboard lets the user choose the strategy, random seed and simulation speed.	Should
FR-15	Reports show water use per day, plant and run, and compare the three strategies on water used, time in stress and final growth.	Must
FR-16	The user can export report data as CSV.	Could

3 Non-Functional Requirements

Numeric targets are proposed values and can be adjusted.

ID	Category	Requirement
NFR-01	Performance	Dashboard pages load within 2 seconds on a normal laptop with 6 months of simulated data.
NFR-02	Performance	Simulating 6 months for 5 plants completes in under 1 minute at full speed.
NFR-03	Safety	The system fails safe: on a sensor fault or missing reading it must not water.
NFR-04	Reproducibility	The same random seed and settings produce identical results.
NFR-05	Data integrity	Foreign keys are enforced, and CHECK and UNIQUE constraints prevent invalid or duplicate rows.
NFR-06	Accuracy	The prediction model has a lower MAE than the naive “same as now” baseline on the test period.

ID	Category	Requirement
NFR-07	Maintainability	Code is split into modules (simulator, engine, db, ml, dashboard), with unit tests for the decision engine and safety rules.
NFR-08	Portability	Sensor input and pump output sit behind `SensorSource` and `PumpController` interfaces, so ESP32 hardware can replace the simulator without changing the engine.
NFR-09	Usability	The dashboard is readable on desktop and phone browsers and uses clear status colours.
NFR-10	Scalability	Indexes on `(run_id, plant_id, timestamp)` keep queries fast with millions of rows.
NFR-11	Security	The system runs locally and uses parameterised SQL queries and input validation to prevent SQL injection.
NFR-12	Compatibility	Python 3.11 or later, SQLite 3, and current versions of Chrome, Firefox and Edge.

4 Database Design

4.1 Overview

The database is SQLite with 11 tables in six groups.

- **Configuration:** `plant_profiles`, `plants`
- **Experiments:** `simulation_runs`
- **Time-series:** `weather`, `sensor_readings`, `watering_events`, `tank_levels`
- **Growth:** `growth_records`
- **AI:** `ml_models`, `predictions`
- **Monitoring:** `alerts`

Conventions: timestamps are ISO-8601 text in UTC, water is measured in ml and moisture in percent. Foreign keys are enabled with `PRAGMA foreign_keys = ON`.

4.2 ER Diagram

Figure 1 shows the tables with their primary keys (PK) and foreign keys (FK). Every `simulation_runs` row is the hub that ties the time-series tables to one experiment run.

4.3 Relationships

All relationships are one-to-many. The schema has no direct one-to-one relationship. Plants and simulation runs are related many-to-many, which is resolved through the log tables (`sensor_readings`, `watering_events`, `growth_records`, `predictions`); each carries both a `run_id` and a `plant_id`.

4.4 Data Dictionary

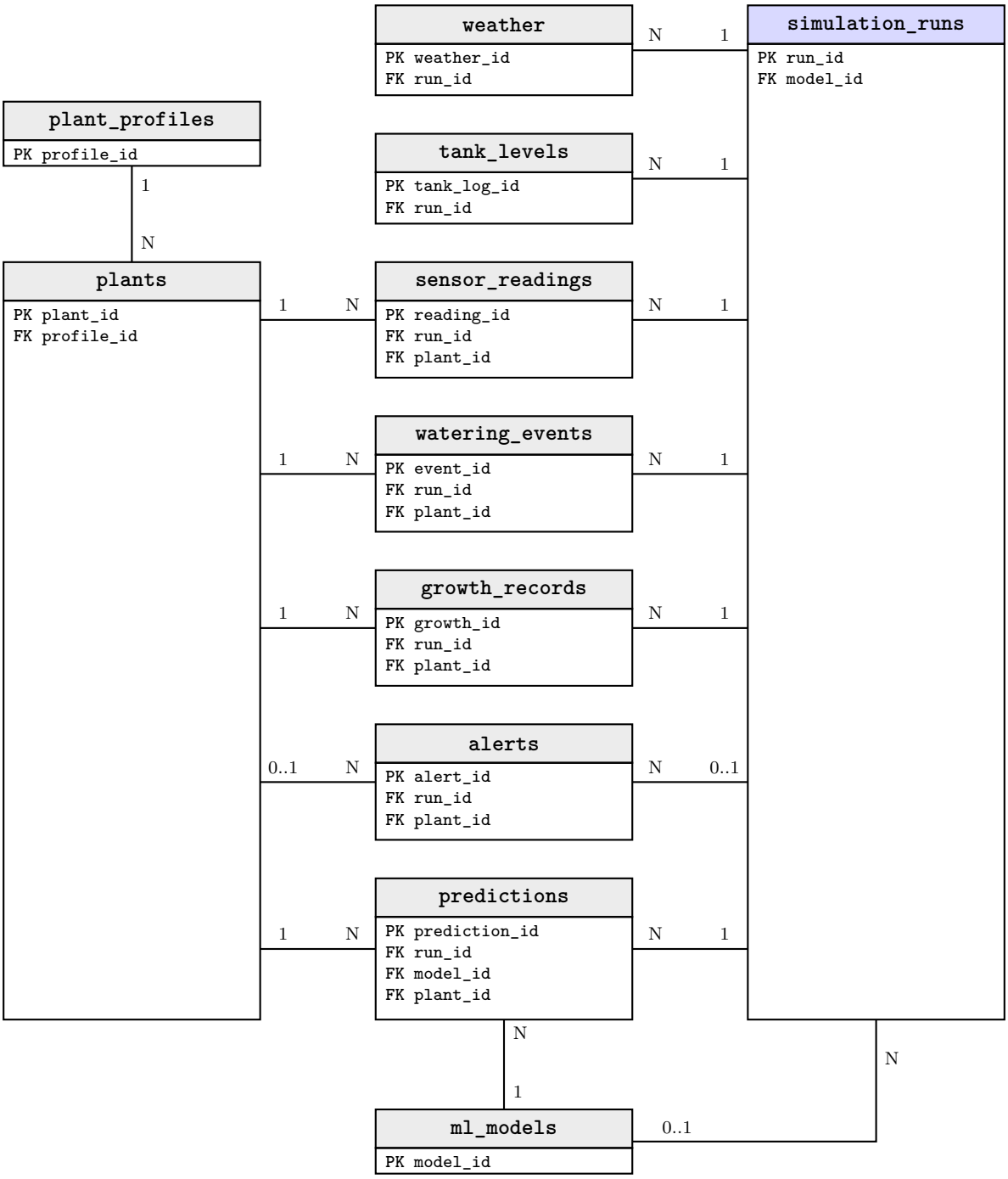


Figure 1: ER diagram of the waterX database. **1** = exactly one, **N** = many (zero or more), **0..1** = optional (nullable foreign key). The line goes from the parent (one) side to the child (many) side.

Parent (1)	Child (N)	Foreign key	Optional	Meaning
plant_profiles	plants	profile_id	No	A profile defines many plants
plants	sensor_readings	plant_id	No	A plant has many readings
plants	watering_events	plant_id	No	A plant receives many waterings
plants	growth_records	plant_id	No	A plant has many growth records
plants	predictions	plant_id	No	A plant has many predictions
plants	alerts	plant_id	Yes	Some alerts (such as low tank) are not tied to a plant
simulation_runs	weather	run_id	No	A run contains many weather records
simulation_runs	sensor_readings	run_id	No	A run contains many readings
simulation_runs	watering_events	run_id	No	A run contains many waterings
simulation_runs	tank_levels	run_id	No	A run contains many tank logs
simulation_runs	growth_records	run_id	No	A run contains many growth records
simulation_runs	predictions	run_id	No	A run contains many predictions
simulation_runs	alerts	run_id	Yes	An alert may be outside any run
ml_models	simulation_runs	model_id	Yes	Only AI-assisted runs use a model
ml_models	predictions	model_id	No	A model makes many predictions

Table 3: Foreign-key relationships.

plant_profiles

Reusable settings for a plant type, such as basil or tomato.

Column	Type	Constraints	Description
profile_id	INTEGER	PK, autoincrement	Unique profile id
name	TEXT	NOT NULL, UNIQUE	Plant type name
min_moisture	REAL	NOT NULL, 0–100	Lower moisture limit (%)
max_moisture	REAL	NOT NULL, 0–100	Upper moisture limit (%); must exceed min_moisture
optimal_moisture	REAL	NOT NULL	Target moisture (%)
water_per_cycle_ml	REAL	NOT NULL, > 0	Water delivered per watering (ml)
max_cycles_per_day	INTEGER	NOT NULL, default 3	Safety limit on waterings per day
max_pump_seconds	INTEGER	NOT NULL, default 30	Safety limit on pump run time
schedule_time	TEXT	nullable	Fixed-schedule time, such as 07:00
uptake_rate	REAL	NOT NULL	Simulator base water-use rate

plants

One plant instance that follows a profile.

Column	Type	Constraints	Description
plant_id	INTEGER	PK, autoincrement	Unique plant id
profile_id	INTEGER	FK to plant_profiles, NOT NULL	The profile this plant follows
name	TEXT	NOT NULL	Plant name
location	TEXT	nullable	Where the plant is
planted_at	TEXT	NOT NULL	Planting date
active	INTEGER	NOT NULL, default 1	1 = active, 0 = deactivated

simulation_runs

One row per simulation run. Same seed with a different strategy gives a fair comparison.

Column	Type	Constraints	Description
run_id	INTEGER	PK, autoincrement	Unique run id
strategy	TEXT	NOT NULL, CHECK	fixed_schedule, rule_based or ai_assisted
random_seed	INTEGER	NOT NULL	Seed for weather and noise
start_time	TEXT	NOT NULL	Simulation start
end_time	TEXT	NOT NULL	Simulation end
model_id	INTEGER	FK to ml_models, nullable	Model used (AI strategy only)
notes	TEXT	nullable	Free text
created_at	TEXT	NOT NULL, default now	Row creation time

ml_models

A trained prediction model and its accuracy.

Column	Type	Constraints	Description
model_id	INTEGER	PK, autoincrement	Unique model id
name	TEXT	NOT NULL	Model name, such as random_forest_v1
algorithm	TEXT	NOT NULL	Algorithm used
horizon_hours	INTEGER	NOT NULL	How many hours ahead it predicts
features	TEXT	NOT NULL	JSON list of feature names
train_start, train_end	TEXT	nullable	Training period
mae, rmse	REAL	nullable	Test-set error metrics
file_path	TEXT	nullable	Saved model file
created_at	TEXT	NOT NULL, default now	Row creation time

`weather`

Simulated weather for each run.

Column	Type	Constraints	Description
`weather_id`	INTEGER	PK, autoincrement	Unique id
`run_id`	INTEGER	FK to `simulation_runs`, NOT NULL, cascade delete	Owning run
`timestamp`	TEXT	NOT NULL	Time of the record
`temperature_c`	REAL	nullable	Temperature (°C)
`humidity_pct`	REAL	nullable	Humidity (%)
`sunlight`	REAL	nullable	Sunlight level
`rain_mm`	REAL	default 0	Rainfall (mm)
UNIQUE (`run_id`, `timestamp`)			

`sensor_readings`

Moisture readings per plant, as the sensor reports them.

Column	Type	Constraints	Description
`reading_id`	INTEGER	PK, autoincrement	Unique id
`run_id`	INTEGER	FK, NOT NULL, cascade delete	Owning run
`plant_id`	INTEGER	FK to `plants`, NOT NULL	Plant measured
`timestamp`	TEXT	NOT NULL	Time of the reading
`moisture_pct`	REAL	nullable	Noisy sensor value (%)
`true_moisture`	REAL	nullable	Simulator ground truth (NULL on real hardware)
`is_fault`	INTEGER	NOT NULL, default 0	1 = dropout or spike
UNIQUE (`run_id`, `plant_id`, `timestamp`)			

`watering_events`

Every time the pump waters a plant.

Column	Type	Constraints	Description
`event_id`	INTEGER	PK, autoincrement	Unique id
`run_id`	INTEGER	FK, NOT NULL, cascade delete	Owning run
`plant_id`	INTEGER	FK, NOT NULL	Plant watered
`timestamp`	TEXT	NOT NULL	Time of watering
`amount_ml`	REAL	NOT NULL, ≥ 0	Water delivered (ml)
`duration_s`	REAL	nullable	Pump run time (s)
`trigger`	TEXT	NOT NULL, CHECK	`schedule`, `threshold`, `prediction` or `manual`
`moisture_before`	REAL	nullable	Moisture before watering
`predicted_moisture`	REAL	nullable	Prediction that triggered it (AI strategy)

`tank_levels`

Water tank level log.

Column	Type	Constraints	Description
`tank_log_id`	INTEGER	PK, autoincrement	Unique id
`run_id`	INTEGER	FK, NOT NULL, cascade delete	Owning run
`timestamp`	TEXT	NOT NULL	Time of the log
`level_ml`	REAL	NOT NULL	Water left (ml)
`capacity_ml`	REAL	NOT NULL	Tank capacity (ml)
UNIQUE (`run_id`, `timestamp`)			

`growth_records`

Plant growth over time.

Column	Type	Constraints	Description
`growth_id`	INTEGER	PK, autoincrement	Unique id
`run_id`	INTEGER	FK, NOT NULL, cascade delete	Owning run
`plant_id`	INTEGER	FK, NOT NULL	Plant measured
`timestamp`	TEXT	NOT NULL	Time of the record
`height_cm`	REAL	nullable	Height (cm)
`leaf_count`	INTEGER	nullable	Number of leaves
`biomass_g`	REAL	nullable	Biomass (g)
`stress_level`	REAL	nullable	0 (none) to 1 (severe)
UNIQUE (`run_id`, `plant_id`, `timestamp`)			

`predictions`

Predicted moisture, stored next to the actual value for evaluation.

Column	Type	Constraints	Description
`prediction_id`	INTEGER	PK, autoincrement	Unique id
`run_id`	INTEGER	FK, NOT NULL, cascade delete	Owning run
`model_id`	INTEGER	FK to `ml_models`, NOT NULL	Model that predicted
`plant_id`	INTEGER	FK, NOT NULL	Plant predicted
`made_at`	TEXT	NOT NULL	When the prediction was made
`target_time`	TEXT	NOT NULL	Time being predicted
`predicted_moisture`	REAL	NOT NULL	Predicted moisture (%)
`actual_moisture`	REAL	nullable	Filled in later

alerts

Warnings raised by the system.

Column	Type	Constraints	Description
alert_id	INTEGER	PK, autoincrement	Unique id
run_id	INTEGER	FK, nullable, cascade delete	Related run
plant_id	INTEGER	FK, nullable	Related plant
timestamp	TEXT	NOT NULL	Time of the alert
type	TEXT	NOT NULL, CHECK	low_tank, too_dry, too_wet, sensor_fault, pump_limit or prediction_dry
severity	TEXT	NOT NULL, CHECK	info, warning or critical
message	TEXT	nullable	Description
resolved	INTEGER	NOT NULL, default 0	1 = resolved

4.5 Indexes and Views

Time-series queries filter by run, plant and time, so these indexes exist:

- sensor_readings(run_id, plant_id, timestamp)
- watering_events(run_id, plant_id, timestamp)
- growth_records(run_id, plant_id, timestamp)
- predictions(run_id, plant_id, target_time)
- alerts(run_id, timestamp)

The view daily_water_usage sums waterings and water use per run, plant and day. It feeds the reports and the strategy comparison.